

## MANAUS GOMES, Brazil

WMO: 821110

Lat: 3.039S

Lon: 60.050W

Elev: 81

StdP: 100.36

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 4             | 21.8       | 21.9 | 18.0                          | 13.1 | 32.7 | 19.1 | 14.0 | 31.3 | 7.0                   | 26.6 | 5.9 | 28.3 | 0.8                   | 350  | 0.291 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 10            | 9.5                    | 35.8            | 26.0 | 35.0 | 25.7 | 34.1 | 25.7 | 28.2                | 32.2 | 27.7 | 31.6 | 27.3 | 31.2 | 3.4                  | 90   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 27.2                            | 23.2 | 29.3 | 27.0 | 23.0 | 29.1 | 26.4 | 22.1 | 28.5 | 91.2          | 32.6 | 89.3 | 32.0 | 87.1 | 31.4 | 30.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 6.2           | 5.3  | 4.6 | DB                         | 20.8 | 36.8               | 0.9 | 1.1                                                | 20.2 | 37.6       | 19.7 | 38.2       | 19.2 | 38.8       | 18.6 | 39.6 | (4) |
| (5)               |      |     | WB                         | 19.9 | 28.5               | 1.1 | 1.2                                                | 19.1 | 29.4       | 18.5 | 30.1       | 17.9 | 30.8       | 17.1 | 31.7 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan    | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)    | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 27.6   | 27.1  | 27.0  | 27.1  | 27.0  | 27.1  | 27.3  | 27.6  | 28.2  | 28.4  | 28.5  | 28.1  | 27.4 | (6)  |
| (7)  |                                                                                  | DBStd       | 1.48   | 1.51  | 1.41  | 1.32  | 1.29  | 1.20  | 1.31  | 1.25  | 1.24  | 1.46  | 1.55  | 1.43  | 1.47 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (8)  |
| (9)  |                                                                                  | HDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (9)  |
| (10) |                                                                                  | CDD10.0     | 6418   | 529   | 477   | 530   | 511   | 531   | 520   | 546   | 565   | 553   | 574   | 543   | 539  | (10) |
| (11) |                                                                                  | CDD18.3     | 3376   | 271   | 243   | 272   | 261   | 273   | 270   | 287   | 307   | 303   | 316   | 293   | 281  | (11) |
| (12) |                                                                                  | CDH23.3     | 30578  | 2219  | 1947  | 2153  | 2063  | 2246  | 2403  | 2704  | 3128  | 3134  | 3285  | 2841  | 2456 | (12) |
| (13) | CDH26.7                                                                          | 12044       | 740    | 613   | 671   | 650   | 726   | 930   | 1154  | 1474  | 1486  | 1528  | 1189  | 883   | (13) |      |
| (14) | Wind                                                                             | WSAvg       | 1.5    | 1.6   | 1.6   | 1.6   | 1.4   | 1.3   | 1.5   | 1.6   | 1.5   | 1.6   | 1.5   | 1.4   | (14) |      |
| (15) | Precipitation                                                                    | PrecAvg     | 2288   | 284   | 285   | 326   | 319   | 243   | 116   | 76    | 56    | 73    | 115   | 182   | 238  | (15) |
| (16) |                                                                                  | PrecMax     | 3156   | 570   | 619   | 633   | 535   | 555   | 228   | 205   | 141   | 243   | 253   | 337   | 519  | (16) |
| (17) |                                                                                  | PrecMin     | 1764   | 33    | 108   | 134   | 136   | 66    | 10    | 3     | 2     | 0     | 11    | 13    | 90   | (17) |
| (18) |                                                                                  | PrecStd     | 292    | 90    | 110   | 108   | 98    | 98    | 47    | 47    | 40    | 52    | 62    | 86    | 101  | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 34.2  | 34.1  | 33.9  | 33.9  | 33.2  | 34.2  | 34.9  | 36.0  | 36.9  | 37.0  | 35.8  | 35.1 | (19) |
| (20) |                                                                                  |             | MCWB   | 26.3  | 26.4  | 27.3  | 27.4  | 26.6  | 26.8  | 26.1  | 26.3  | 26.0  | 26.4  | 26.4  | 26.5 | (20) |
| (21) |                                                                                  | 2%          | DB     | 33.0  | 32.9  | 32.8  | 32.2  | 32.2  | 33.1  | 33.9  | 35.0  | 35.6  | 35.2  | 34.2  | 33.8 | (21) |
| (22) |                                                                                  |             | MCWB   | 26.1  | 26.0  | 26.4  | 26.2  | 26.1  | 26.0  | 25.7  | 25.5  | 25.5  | 25.7  | 26.0  | 26.1 | (22) |
| (23) |                                                                                  | 5%          | DB     | 31.9  | 31.8  | 31.8  | 31.8  | 31.8  | 32.1  | 33.0  | 34.0  | 34.8  | 34.5  | 33.8  | 32.2 | (23) |
| (24) |                                                                                  |             | MCWB   | 25.9  | 25.8  | 26.0  | 26.0  | 26.0  | 25.2  | 25.1  | 25.1  | 25.3  | 25.6  | 25.9  | 25.9 | (24) |
| (25) |                                                                                  | 10%         | DB     | 30.8  | 30.2  | 30.2  | 30.2  | 30.8  | 31.2  | 32.0  | 33.1  | 33.2  | 33.2  | 32.2  | 31.1 | (25) |
| (26) |                                                                                  |             | MCWB   | 25.7  | 25.6  | 25.7  | 25.7  | 25.9  | 25.2  | 24.9  | 24.9  | 25.3  | 25.7  | 25.8  | 25.9 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 28.0  | 28.2  | 28.3  | 28.5  | 28.2  | 28.2  | 27.9  | 28.1  | 28.1  | 28.4  | 28.2  | 28.2 | (27) |
| (28) |                                                                                  |             | MCD B  | 31.1  | 31.5  | 31.8  | 31.8  | 31.4  | 32.1  | 31.9  | 33.0  | 33.7  | 33.7  | 32.4  | 31.9 | (28) |
| (29) |                                                                                  | 2%          | WB     | 27.2  | 27.2  | 27.3  | 27.4  | 27.4  | 27.2  | 27.1  | 27.2  | 27.4  | 27.6  | 27.5  | 27.4 | (29) |
| (30) |                                                                                  |             | MCD B  | 30.4  | 30.2  | 30.5  | 30.3  | 30.3  | 30.7  | 31.3  | 32.1  | 32.6  | 32.3  | 31.7  | 30.9 | (30) |
| (31) |                                                                                  | 5%          | WB     | 26.7  | 26.6  | 26.7  | 26.7  | 26.8  | 26.7  | 26.6  | 26.7  | 27.0  | 27.1  | 27.0  | 26.9 | (31) |
| (32) |                                                                                  |             | MCD B  | 29.6  | 29.3  | 29.5  | 29.4  | 29.5  | 29.9  | 30.3  | 31.2  | 31.7  | 31.6  | 31.2  | 30.2 | (32) |
| (33) |                                                                                  | 10%         | WB     | 26.2  | 26.1  | 26.2  | 26.2  | 26.2  | 26.1  | 26.1  | 26.2  | 26.4  | 26.5  | 26.5  | 26.4 | (33) |
| (34) |                                                                                  |             | MCD B  | 28.7  | 28.5  | 28.5  | 28.5  | 28.7  | 28.9  | 29.3  | 30.1  | 30.4  | 30.6  | 30.1  | 29.1 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 7.3   | 7.1   | 7.1   | 7.2   | 7.3   | 8.2   | 9.0   | 9.9   | 9.5   | 8.7   | 7.7   | (35) |      |
| (36) |                                                                                  |             | MCDBR  | 8.9   | 8.9   | 8.8   | 8.9   | 8.8   | 9.3   | 10.2  | 11.1  | 11.5  | 11.1  | 10.3  | 9.4  | (36) |
| (37) |                                                                                  |             | MCWBR  | 3.4   | 3.4   | 3.5   | 3.5   | 3.5   | 3.4   | 3.4   | 3.4   | 3.3   | 3.3   | 3.4   | 3.5  | (37) |
| (38) |                                                                                  | 5% WB       | MCD BR | 7.9   | 7.7   | 8.0   | 7.9   | 7.9   | 8.6   | 9.3   | 10.2  | 10.4  | 10.1  | 9.2   | 8.4  | (38) |
| (39) |                                                                                  |             | MCWBR  | 3.6   | 3.7   | 3.7   | 3.7   | 3.6   | 3.8   | 3.7   | 3.9   | 3.8   | 3.7   | 3.7   | 3.8  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.399  | 0.394 | 0.393 | 0.385 | 0.387 | 0.391 | 0.403 | 0.486 | 0.526 | 0.505 | 0.492 | 0.451 | (40) |      |
| (41) |                                                                                  | taud        | 2.521  | 2.546 | 2.553 | 2.586 | 2.569 | 2.558 | 2.502 | 2.227 | 2.097 | 2.179 | 2.215 | 2.351 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 935    | 940   | 930   | 915   | 887   | 870   | 865   | 813   | 802   | 830   | 842   | 881   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 111    | 109   | 107   | 100   | 98    | 97    | 104   | 142   | 167   | 155   | 149   | 130   | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 4.31   | 4.26  | 4.29  | 4.20  | 4.14  | 4.59  | 4.92  | 5.30  | 5.30  | 5.06  | 4.67  | 4.20  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.36   | 0.34  | 0.32  | 0.16  | 0.27  | 0.28  | 0.25  | 0.30  | 0.40  | 0.28  | 0.32  | 0.35  | (45) |      |

## Historical Trends

|      |                       | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |      |  |
|------|-----------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|------|--|
|      |                       |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |      |  |
|      |                       | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |      |  |
| (46) | Station Only          | -0.22 | -0.11   | -1.63  | -0.52   | -1.15 | -1.26 | N/A         | N/A     | -153    | -153    | (46) |  |
| (47) | Regional (1 neighbor) | -0.22 | -0.11   | -1.63  | -0.52   | -1.15 | -1.26 | N/A         | N/A     | -153    | -153    | (47) |  |

Nomenclature: See separate page